Structural health monitoring

- Modal characteristics estimation and damage identification of a shear structure using the eigensystem realization algorithm (ERA)
- Designed a min-max, energy-balanced routing tree, and a node deployment strategy to minimize energy consumption

Intelligent parking lot monitoring

- Ultrasonic sensors and magnetometers for accurate vehicle detection
- Verified by experiments conducted in a multi-storied university parking lot

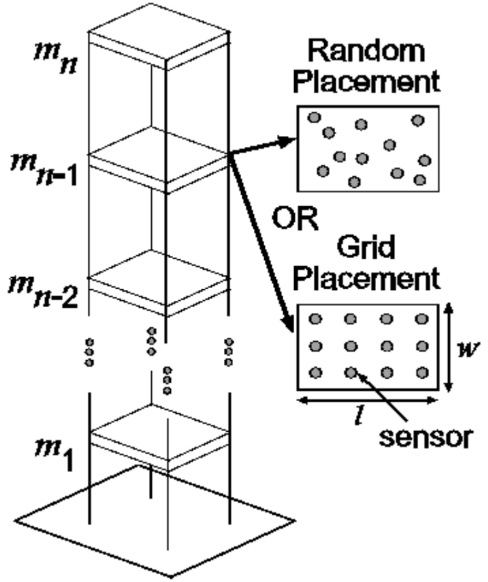


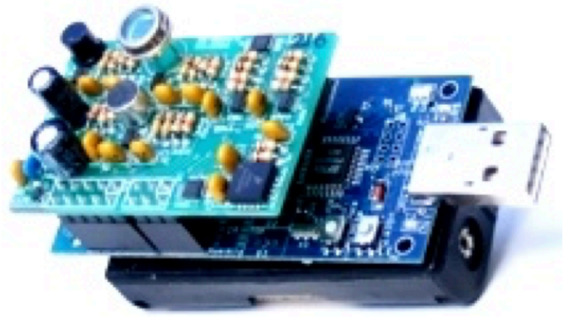
Fig 1. An n-story structure with sensors placed randomly or in a grid on each floor.

Fig 2. (a) SBT80 sensor board mounted on a Tmote Sky.

Tmote Sky is an ultra low power wireless module, featuring a 250 kbps radio, 10 KB RAM, 48 kB flash, and 1 MB storage.

SBT80 has six different sensors: visual light, infrared, temperature, acoustic, magnetometer, and accelerometer.

(b) SRF02 ultrasonic sensors can measure an object's distance by emitting ultrasonic waves and perceiving the reflected waves



(a)

(b)